Soap Film Physics-Based Simulation

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Figure 1: Rendered images of real-time, physics-based soap bubble simulation.

Abstract

We propose a real-time, interactive physics-based simulation framework for soap bubbles. Our model extends Position Based Dynamics (PBD) with a novel surface tension constraint to accurately model macroscopic bubble deformations. Furthermore, we simulate microscopic film thickness evolution as a wave propagation process via the discrete Laplace-Beltrami operator, integrating it directly into the PBD pipeline. The dynamically computed thickness field is then transferred to the shader pipeline to render realistic optical phenomena, including the Fresnel effect and thin-film interference. Our results show that coupling physical governing equations with a real-time solver achieves both highly stable dynamics and visually realistic interferometric effects.

1 Introduction

The trade-off between physical plausibility and real-time feasibility has been a challenge in computer graphics. Our main goal of the project is to simulate the macroscopic motion and microscopic thickness of the soap film real-time and interactive. To achieve this, we use Position Based Dynamics (PBD) with our proposed surface tension constraint to keep the shape of the soap bubble. Furthermore, we implement the thickness evolution pipeline to simulate the film thickness dynamically. By simulating the thickness physics-based, the interferometric effects can be simulated easily in shader.

2 Related Work

2.1 Real-time Simulations

One of the most widely used real-time physics-based methodologies for rigid body or cloth simulation, Position Based Dynamics (PBD)[8], is still being used for the sake of its stable and rapid performance. The problem of methodology that the rigidity of the object changes as the time step varies was solved in Extended Position Based Dynamics (XPBD)[6]. The methodology is still widely used where real-time simulations are needed as an extension of the approach, such as XPBI[10].

There have been studies to apply Position Based Dynamics in soap bubbles such as Position Based Surface Tension Flow[9], while the study focused on the stability of the film to the contrast of the original PBD. Approaches used other than PBD to simulate soap bubbles[2], but the trade-off between physical plausibility and real-time feasibility have been a challenge in computer graphics.

2.2 Soap Film Dynamics

Soap bubbles were first statistically modeled in computer graphics[4]. The study modeled the static shape and thin film interference. The dynamics of thin films were studied and modeled as a gradient flow model and discretized to simulate in triangle meshes[1]. The most recent study[5] proposed a model that simulates evolving thickness accurately in the use of Navier-Stokes equation. The accurate physics based model enabled a simulation to show beautiful and realistic inteferometric effects.

3 Position Based Dynamics

Our model uses Position Based Dynamics[8] to simulate real-time and interactive. PBD uses physical constraints to update each points of object's mesh every timestep, the positions are predicted using Verlet integration, then the exact positions are determined by the constraints in the use of Gauss-Seidel method. For constraint C as a function of positions $\mathbf{p}$, The position update is approximated as the following.

$$C(\mathbf{p} + \Delta\mathbf{p}) \approx C(\mathbf{p}) + \nabla_{\mathbf{p}} C(\mathbf{p}) \cdot \Delta\mathbf{p} = 0 \tag{1}$$

Which leads to the position update rule:

$$\Delta\mathbf{p} = -\frac{C(\mathbf{p})}{|\nabla_{\mathbf{p}} C(\mathbf{p})|^2} \nabla_{\mathbf{p}} C(\mathbf{p}) \tag{2}$$

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We used two main constraints, volumetric constraint and surface tension constraint to simulate realistic soap film dynamics.

3.1 Volumetric Constraint

Fundamentally, stretching constraint is used to guarantee the connectivity of the mesh. The function keeps the length of the edges.

$$C_{stretch}(\mathbf{p}_i, \mathbf{p}_j) = |\mathbf{p}_i - \mathbf{p}_j| - l_0 \tag{3}$$

The main constraint that keeps the shape of the soap bubble is the volumetric constraint. It keeps the volume of the object constant. By adjusting $k_{pressure}$, the volume of the soap bubble can be controlled.

$$C_{volume}(p_1, \cdots, p_N) = \left(\sum_{i=1}^{n_{triangles}} \left(p_{t_1^i} \times p_{t_2^i} \right) \cdot p_{t_3^i} \right) - k_{pressure} V_0 \tag{4}$$

For numerical stability, we used the idea from XPBD[6]. The position is updated with compliance α, which makes the bubble incompressible when it gets closer to 0.

$$\lambda = \frac{-C_{volume}}{\sum_i |\nabla_{\mathbf{p}_i} C_{volume}|^2 + \alpha} \tag{5}$$

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \lambda \nabla_{\mathbf{p}_i} V \tag{6}$$

3.2 Surface Tension Constraint

Following the method of original PBD, bending constraint that keeps the angle between the faces were implemented. However, implemented object showed rubber-like motion than a soap bubble. Instead of the bending, we considered the surface tension as a physical constraint. The surface tension is a force that minimizes surface free energy, which is a gradient of a surface area.

$$F_i = -\nabla_{\mathbf{p}_i} E = -\gamma \nabla_{\mathbf{p}_i} A \tag{7}$$

The constraint can be set as $C_{surface}(\mathbf{p}) = A_i = 0$. Calculating the area around the point $\mathbf{p}_i$ as in **Figure 2**, the gradient can be derived using the angle of opposite triangles as in equation (8). Using our proposed physics based constraint let the simulation show more bubble-like motion that restores its shape smoothly.

$$\nabla_{\mathbf{p}_i} C_{surface}(\mathbf{p}_i, \mathbf{p}_j) = \frac{1}{2} \sum_{j \in N(i)} (\cot \alpha_{ij} + \cot \beta_{ij})(\mathbf{p}_i - \mathbf{p}_j) \tag{8}$$

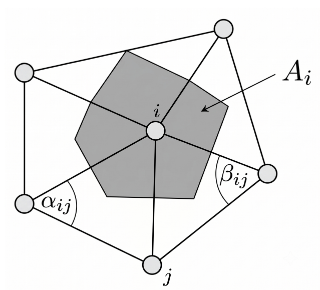

Figure 2: Mesh diagram for surface tension constraint.

When implementing the constraint to the PBD pipeline, we applied the gradient as an explicit acceleration. When it is used as hard constraint $C_{surface}(\mathbf{p}) = 0$, the position update $\Delta \mathbf{p}_i = -\lambda \nabla_{\mathbf{p}_i} C$ can cause a conflict with volumetric constraint that the bubble shrinks and the system diverges.

4 Thin-Film Dynamics

Our main goal of the project is to simulate the thickness of the soap film dynamically. The simulated thickness can be transferred to the shader pipeline that renders interferometric effects such as interference. We used the idea of considering the film thickness as a wave propagation and implementing to the PBD pipeline.

4.1 Laplace-Beltrami Operator

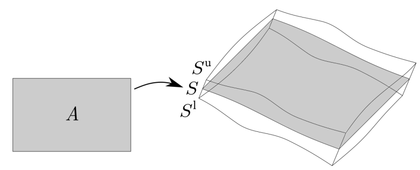

Figure 3: Soap film geometry modeled as a smooth map.

The temporal evolving thickness of soap film was studied recently in this paper[5]. The paper represents soap film geometry as a smooth map

$$S : A \rightarrow \mathbb{R} \tag{9}$$

and define upper and lower boundary, where h is the thickness field and $\mathbf{n}$ is the normal of the film.

$$S^u := S + \frac{1}{2} h\mathbf{n} \tag{10}$$

$$S^l := S - \frac{1}{2} h\mathbf{n} \tag{11}$$

Starting from Navier-Stokes equation:

$$\rho \frac{D\dot{\mathbf{x}}}{Dt} = -2\sigma(\delta_{S^u} H^u \mathbf{n}^u + \delta_{S^l} H^l \mathbf{n}^l) - \nabla p + f \tag{12}$$

integrating over the domain $(-h/2, h/2)$ and some technical integration[5] results the following.

$$\frac{D\dot{h}}{Dt} = \frac{1}{\rho}(\sigma \nabla_s^2 h - Q) \tag{13}$$

The pressure difference Q between upper and lower boundary can be approximated to 0 since it does not affect the thickness evolution.

We now know that the dynamics of the thickness field h can be considered as a wave propagation. The continuous governing equation can be discretized through discrete Laplace-Beltrami Operator.

$$\nabla^2 h_i = \sum_{j \in N(i)} (h_j - h_i) \tag{14}$$

Considering the temporal damping, the final governing equation is derived as equation (15), for constants c^2 and k_{damp}.

$$\frac{\partial^2 h_i}{\partial t^2} = c^2 \nabla^2 h_i - k_{damp} \frac{\partial h_i}{\partial t} \tag{15}$$

4.2 Pipeline Implementation

The key idea of implementing thickness evolution real time is to implement the thickness field h into the PBD pipeline. Corresponding the thickness to position and the time derivative to velocity, it follows the same logic to set constraints to the original Position Based Dynamics.

It follows the steps: prediction, constraint projection, and velocity update. The first step is to predict the next step using film velocity v_i.

$$h_i^* = h_i^t + \Delta t \cdot v_i^* \tag{16}$$

The constraint is to keep the total amount of soap liquid constant to the initial value V_{init}.

$$C(h_1, \cdots, h_N) = \sum_i (A_i \cdot h_i) - V_{init} = 0 \tag{17}$$

The film velocity is updated using the governing equation.

$$v_i^{new} = v_i^{old}(1 - k_{damp}) + c^2 \nabla^2 h_i \Delta t \tag{18}$$

The thickness evolution pipeline is shown in **Algorithm 1**.

Algorithm 1 Thickness evolution pipeline

for all faces i **do**
  initialize $h_i = h_i^0, v_i = v_i^0$
end for
loop
  for all faces i **do** $h_i \leftarrow h_i + \Delta t v_i$
  loop solverIterations **times**
    projectConstraints$(C_1, \ldots, C_M, h_1, \ldots, h_N)$
  endloop
  for all faces i **do**
    $v_i \leftarrow v_i(1 - k_{damp}) + \Delta t c^2 \nabla^2 h_i$
  end for
  velocityUpdate$(v_1, \ldots, v_N)$
end loop

5 Interferometric Effects

The last goal of the project is to implement realistic optical effects of soap bubbles. In the use of dynamically simulated thickness field, the interferometric effects can be simulated easily in shader with equations and precomputed look-up table.

5.1 Fresnel Effect

To simulate the Fresnel effect, which an optical effect where the reflectivity changes depending on the viewing angle. Transparent spheres show higher reflectance as it approaches the edge. Implementing the effect will make the bubble realistic. Schlick's approximation is used to calculate the reflectance[7].

$$R(\theta) = R_0 + (1 - R_0)(1 - \cos\theta)^5 \tag{19}$$

where

$$R_0 = \left(\frac{1-n}{1+n}\right)^2 \tag{20}$$

The implementation of reflectance difference was done through alpha blending in shader level.

5.2 Thin-Film Interference

The thickness of the film is dynamically calculated in PBD pipeline. The beautiful rainbow of the bubble is caused by thin-film interference, which occurs from path difference Δ on the surface:

$$\Delta = 2nh\sqrt{1 - \frac{1 - \cos^2\theta}{n^2}} \tag{21}$$

where n is the refractive index of the soap. We used $n = 1.3$ for the simulation.

We can determine the color with the path difference and refracting angle. For real-time rendering, we used 2D Look-Up Table that returns RGB values as RGB = LUT(Δ, θ) as in **Figure 4**. Look-Up Table is calculated from sun spectrum[3]. The wavelength is extracted from visible light which is between 380-780 nm. Planck function of $T = 5777$K was used, and transfered to XYZ color space. CIE color matching function[3] is used to return RGB value.

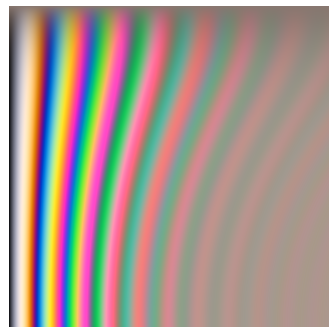

Figure 4: Look-Up Table: horizontal axis - path difference, vertical axis - reflected angle.

6 Results and Discussion

Results of the rendered simulation is shown in **Figure 1** and **Figure 5**. By simulating the dynamic thickness of the soap bubble, we can see that interferometric effects are visualized realistically.

Figure 5: Realistically rendered bubble.

The key point of the project was to simulate physics based. The real world is governed by physics, so that rendering based on physical governing equation enables the simulator to render realistic results.

7 Conclusion

In this project, we successfully simulated the realistic soap film real-time. Our proposed surface tension constraint let the simulation show more bubble-like motion than traditional bending constraints. Also, implementing the thickness evolution to the PBD pipeline enabled us to calculate the dynamic thickness field for thin-film interference. The significance of the project is that the real world is governed by physics, so rendering based on physical governing equation enables the simulator to render realistic results. Our approach proves that implementing physical constraints and wave propagation is a powerful way to simulate realistic optical effects.

8 Acknowledgments

The project is done as a final project of Graphics Programming, 2026, under Prof. Young Min Kim. Graduate course of Electrical and Computer Engineering, Seoul National University. Source code at https://github.com/torytony24/graphics_project.git.

References

[1] Omri Azencot, Orestis Vantzos, Max Wardetzky, Martin Rumpf, and Mirela Ben-Chen. 2015. Functional thin films on surfaces. In *Proceedings of the 14th ACM SIGGRAPH / Eurographics Symposium on Computer Animation*. ACM, 137–146. doi:10.1145/2786784.2786793

[2] Fang Da, Christopher Batty, Chris Wojtan, and Eitan Grinspun. 2015. Double bubbles sans toil and trouble. *ACM Transactions on Graphics* 34 (2015), 1–9. doi:10.1145/2767003

[3] Amanda Ghassaei. 2026. *SoapFlow: WebGL Soap Film Fluid Simulation*. https://github.com

[4] A. Glassner. 2000. Soap bubbles. 1. *IEEE Computer Graphics and Applications* 20 (2000), 76–84. doi:10.1109/38.865884

[5] Sadashige Ishida, Peter Synak, Fumiya Narita, Toshiya Hachisuka, and Chris Wojtan. 2020. A model for soap film dynamics with evolving thickness. *ACM Transactions on Graphics* 39 (2020). doi:10.1145/3386569.3392405

[6] Miles Macklin, Matthias Müller, and Nuttapong Chentanez. 2016. XPBD. In *Proceedings of the 9th International Conference on Motion in Games*. ACM, 49–54. doi:10.1145/2994258.2994272

[7] Zander Majercik. 2021. *The Schlick Fresnel Approximation*. Apress, 109–114. doi:10.1007/978-1-4842-7185-8_9

[8] Matthias Müller, Bruno Heidelberger, Marcus Hennix, and John Ratcliff. 2007. Position based dynamics. *Journal of Visual Communication and Image Representation* 18 (2007), 109–118. doi:10.1016/j.jvcir.2007.01.005

[9] Jingrui Xing, Liangwang Ruan, Bin Wang, Bo Zhu, and Baoquan Chen. 2022. Position-Based Surface Tension Flow. *ACM Transactions on Graphics* 41 (2022), 1–12. doi:10.1145/3550454.3555476

[10] Chang Yu, Xuan Li, Lei Lan, Yin Yang, and Chenfanfu Jiang. 2024. XPBI: Position-Based Dynamics with Smoothing Kernels Handles Continuum Inelasticity. In *SIGGRAPH Asia 2024 Conference Papers*. ACM, 1–12. doi:10.1145/3680528.3687577